SS-P-821b
August 17, 1966

SUPERSEDING
Int. Fed. Spec. SS-P-00821a (GSA-FSS)
March 13, 1962 and
Fed. Spec. SS-P-821
October 8, 1948

FEDERAL SPECIFICATION

PUMICE; GROUND, ABRASIVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers one type of ground abrasive pumice for general polishing and finishing use.

1.2 **Classification**

1.2.1 *Grades.* Ground pumice shall be of the grades shown in table I, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

RR-S-366—Sieve, Test.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale an a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-3816—Abrasives and Abrasive Products, for Shipment and Storage, Packaging and Packing of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** The pumice shall be a ground product resulting from the processing of natural deposits.

3.1.1 *Excluded impurities.* The ground pumice shall be free from quartz or flint, and shall not contain more than 5 percent of obsidian and similar hard material (see 4.4.2).

3.1.2 *Abrasive action.* The ground pumice shall be furnished dry and shall have an abrasive action that will polish but not scratch glass (see 4.4.1).

3.1.3 *Fineness.* The ground pumice shall be of the fineness shown in table I for the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:14Z
Check the source to verify that this is the current version before use.

respective grades, all percentages being by weight, the sieves used shall conform to the requirements of RR-S-366 (see 4.4.3).

TABLE I

Grade		Fineness
	Percent	
2	95 through No. 30	Sieve, on No. 80
1-1/2	95 through No. 40	Sieve, on No. 100
1	95 through No. 40	Sieve, on No. 120
1/2	95 through No. 50	Sieve, on No. 140
0-3/4	95 through No. 60	Sieve, on No. 200
		Percent
0-1/2	98 through No. 80	Sieve, and 31 through No. 325
0	98 through No. 120	Sieve, and 40 through No. 325
F	98 through No. 100	Sieve, and 63 through No. 325
FF	98 through No. 140	Sieve, and 75 through No. 325
FFF	98 through No. 200	Sieve, and 83 through No. 325
FFFF	98 through No. 200	Sieve, and 90 through No. 325

3.1.4 *Tolerance.* For grades F to FFFF inclusive, the indicated percentage required to pass a No. 325 sieve minus 10 percent will be acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for acceptance.

4.2.1 *Inspection lot.* For purposes of sampling, a lot shall consist of all containers of the same grade and capacity offered for delivery at one time. The sample unit shall be one filled container.

4.2.2 *Sampling for inspection of filled containers.* A random sample of filled containers shall be taken from each lot in accordance with MIL-STD-105, at inspection level I and acceptable quality level (AQL) of 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.2.3 *Sampling for tests.* A grab sample of not less than 1 ounce shall be taken from not less than 1 percent of the vendor's shipping containers, provided such containers contain not less than 100 pounds (45.4 kg). In case of smaller containers, a grab sample of not less than 1 ounce shall be taken at random from each lot of containers totaling not more than 5,000 pounds, or fraction thereof. The gross sample shall in all cases consist of not less than three grab portions taken at random from separate containers. With very large lots where the sample drawn as above will amount to more than 10 pounds, the percentage of packages sampled shall be reduced so that the amount drawn shall not exceed 10 pounds. The inspector shall rapidly mix the gross sample and place it in an airtight container, which shall be filled, sealed, marked, accurately weighed, its weight and date of weighing recorded on the package, and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.3 Inspection.

4.3.1 *Inspection of filled container.* Each sample filled container selected in accord-

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:14Z
Check the source to verify that this is the current version before use.

ance with 4.2.2 shall be examined for defects of the container and closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 *Lot acceptance tests.* The gross sample specimen selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.4. If the specimen fails in one or more of these tests, the lot shall be rejected.

4.4 **Tests.**

4.4.1 *Scratch test.* A scratch test shall be made of each sample by placing a small quantity on a clean piece of plate glass and then rubbing the powder over the surface of the glass. Any pronounced scratches produced will be cause for rejection of the entire shipment.

4.4.2 *Impurities test.* A petrographic analysis will be made at the option of the inspector to determine impurities as listed in 3.1.1.

4.4.3 *Fineness test.* The sieves used shall be stacked one below the other in order of fineness, with a pan attached below the finest sieve. Place a 50 gram example on the top sieve, cover, and set the nest of sieves on the holder of a sieve machine. Run for 30 minutes on a machine geared to give 285 gyrations and 152 taps of the striker per minute. Weigh the amount left on each sieve and pan separately and calculate the percentage. The sieving machine may be any mechanical shaker that gives the above rate of operation.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.** The abrasive shall be packaged and packed in accordance with MIL-P-3816, as specified for the applicable level (see 6.2).

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to any special markings specified in the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Pumice conforming to this specification is intended for polishing metal, plastic, hard rubber parts, finishing glass, mirrors, and furniture.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Grade required (see 1.2.1).
(*c*) Selection of applicable levels of packaging and packing required (see 5.1).

MILITARY CUSTODIANS:

Army—MR
Navy—Ships
Air Force—69

Review activities:

Army—MR, GL, MO
Air Force—69

User activity:

Navy—Ships, YD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:14Z
Check the source to verify that this is the current version before use.

MILITARY COORDINATING ACTIVITY:
Army—MR

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:14Z
Check the source to verify that this is the current version before use.